

The Ramond $\text{Vir} \oplus \text{Vir}$ Branching Coefficient: Direct Screening Matrix Elements and PBW Checks

Machine note

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Abstract

We formulate the Ramond analogue of the Hadasz–Jaskolski blow-up factor as the matrix element of an explicit double-*Virasoro* primary in the free field. The two Ramond spin fields turn its fermionic part into a zero-mode-resolved Pfaffian. For the first crossed state $(0, \frac{3}{4}, \frac{3}{4})$ we evaluate this matrix element directly on the two- and three-screening neutrality planes. The two-screening Selberg integral produces the irreducible crossed polynomial H , while the natural three-screening determinant produces the factorized polynomial K . We also retain a generic-momentum PBW/Ward path sum as an independent finite-level oracle. It is not itself a screening derivation. A generic-momentum, all-level Ramond screening formula remains open. The presently proved endpoint residues are rank-one maps on a vector-valued contour problem; completion requires finite contour-orbit closure, a full-rank set of collision and inversion/connection maps, base vectors, and only then same-parity momentum interpolation.

Correction of scope. An earlier version called the PBW/Ward path sum below the final free-field formula. That was too strong: it is an exact basis reconstruction of the three-form, hence an oracle and check, but not an evaluation of the Coulomb matrix element. The genuinely screened results of this version are stated in Section 6.3.

1 Conventions and statement of the problem

We use the conventions of `Human Notes/SCblock.tex`. Put

$$Q = b + b^{-1}, \quad h_1 = \frac{1}{8}Q^2 - \frac{1}{2}P_1^2, \quad h_j = \frac{1}{16} + \frac{1}{8}Q^2 - \frac{1}{2}P_j^2 \quad (j = 2, 3). \quad (1)$$

The ordered trinion is $(\text{NS}, \text{R}, \text{R})$, with the NS state in slot 1. Its branch labels obey

$$n_1 \in \frac{1}{2}\mathbb{Z}, \quad n_2, n_3 \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}. \quad (2)$$

It is useful to put $j_i = 2n_i$. Thus j_1 is integral and j_2, j_3 are half-integral. A Ramond summand occurs twice; we denote its copy by $\epsilon_i \in \{0, 1\}$. The two chiral $(\text{NS}, \text{R}, \text{R})$ forms are $\eta = \pm 1$, and $f \in \{0, 1\}$ denotes the parity of the complete three-form. In the human-note ground basis,

$$\rho_0^{(\eta)} = \rho^{++} + \eta\rho^{--}, \quad \rho_1^{(\eta)} = \rho^{+-} + i\eta\rho^{-+}. \quad (3)$$

The normalized coefficient is

$$\mathbb{B}_f^{(\eta)} = \frac{\hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\epsilon_2}, v_{3,n_3}^{\epsilon_3})}{\|v_{1,n_1}\| \|v_{2,n_2}^{\epsilon_2}\| \|v_{3,n_3}^{\epsilon_3}\|}. \quad (4)$$

Claim (the matrix element and its Selberg realization). Write $\mathbf{n} = (n_1, n_2, n_3)$ and $\epsilon = (\epsilon_2, \epsilon_3)$. The quantity which the free-field calculation has to produce is the raw double-Virasoro-primary matrix element

$$\mathbf{M}_{\epsilon;f}^{(\eta)}(\mathbf{P}; \mathbf{n}) := \langle v_{1,n_1}(P_1) | V_f^{(\eta)} \left[v_{2,n_2}^{\epsilon_2}(P_2) \right] (1) \left| v_{3,n_3}^{\epsilon_3}(P_3) \right\rangle = \hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\epsilon_2}, v_{3,n_3}^{\epsilon_3}). \quad (5)$$

This is not a matrix element of the bare exponential alone: the three states in (5) are the explicit simultaneous $\text{Vir} \oplus \text{Vir}$ primaries, including their consecutive $\chi = f - i\psi$ strings and, on the Ramond legs, the chosen zero-mode components.

Let $\mathbf{g}_{\epsilon_2\epsilon_3;f}^{(\eta)}$ be the raw ground-spin matrix in that component. On a charge-neutral free-field representative with N insertions of the b -screening current, the claim is

$$\mathbf{M}_{\epsilon;f}^{(\eta)} \Big|_N = \mathbf{g}_{\epsilon_2\epsilon_3;f}^{(\eta)} \frac{\mathfrak{S}_N[F_{\mathbf{n},\epsilon}^{(\eta)}; -b\alpha_1 - u, -b\alpha_2 - v, g]}{\mathfrak{S}_N[F_0^{(\eta,f)}; -b\alpha_1 - u_0, -b\alpha_2 - v_0, g]}, \quad g = -\frac{bQ}{2}. \quad (6)$$

Here $\mathfrak{S}_N$ is the Selberg functional (33); $F_{\mathbf{n},\epsilon}$ is the symmetric polynomial obtained by inserting the three branch-state strings in the two-spin Pfaffian, while F_0 is obtained by removing those strings but keeping the same Ramond ground component. Equivalently, after the ground-spin matrix is stripped off, the descendant-to-primary matrix-element ratio is precisely the ratio of the two Selberg integrals in (6). Common ordered-contour phases and factorials cancel in this ratio.

Combining (4) and (6) gives, on that neutrality plane,

$$\mathbb{B}_f^{(\eta)} \Big|_N = \frac{\mathbf{g}_{\epsilon_2\epsilon_3;f}^{(\eta)}}{\|v_{1,n_1}\| \|v_{2,n_2}^{\epsilon_2}\| \|v_{3,n_3}^{\epsilon_3}\|} \frac{\mathfrak{S}_N[F_{\mathbf{n},\epsilon}^{(\eta)}; -b\alpha_1 - u, -b\alpha_2 - v, g]}{\mathfrak{S}_N[F_0^{(\eta,f)}; -b\alpha_1 - u_0, -b\alpha_2 - v_0, g]}. \quad (7)$$

The pole-cleared quantity used later is merely $\mathcal{R}|_N = (\prod_i \delta_{n_i}(P_i)) \mathbf{M}|_N$; the δ factors are not part of the physical three-point matrix element. Also, N here is the number of screening currents, hence the dimension of the integral, not a branch level. A vector of contour-orbit integrals and its projectors may be useful for evaluating the scalar ratio in (6), but that vector is computational auxiliary data rather than an additional branching coefficient. An efficient evaluation algorithm is deferred to the later contour-recursion analysis.

Consequently a function of only (b, P_i, n_i) is not defined in the R sector: one must also say which $(\epsilon_2, \epsilon_3; f, \eta)$ component is meant. Moreover, an unsquared coefficient changes under independent sign choices for the branch highest vectors. We therefore report the root-independent quantity $(\mathbb{B}_f^{(\eta)})^2$; an unsquared representative is fixed by the charge-basis convention below.

One Ramond summand starts at relative conformal weight $2n^2 - 1/16$. The exponent $2n^2 - 1/8$ in the theta-channel sewing formula also subtracts the auxiliary Ramond vacuum weight. These two numbers should not be confused in a level count.

2 The ℓ products and the NS result

For an integer $m > 0$ define

$$\ell(x, m) = \begin{cases} \prod_{\substack{r,s \geq 0, \ r+s < m \\ r+s \equiv 0 \pmod{2}}} (x + rb + sb^{-1}), & m \text{ even,} \\ 2^{1/8} \prod_{\substack{r,s \geq 0, \ r+s < m \\ r+s \equiv 1 \pmod{2}}} (x + rb + sb^{-1}), & m \text{ odd,} \end{cases} \quad (8)$$

and

$$\ell(x, 0) = 1, \quad \ell(x, m) = \begin{cases} (-1)^{-m/2} \ell(Q - x, -m), & m < 0 \text{ even}, \\ \ell(Q - x, -m), & m < 0 \text{ odd}. \end{cases} \quad (9)$$

For later use set

$$D_n(P) = \ell(2P, 4n) \ell(Q + 2P, 4n), \quad \delta_n(P) = 2^{-1_{4n \text{ odd}}/8} \ell(Q + 2P, 4n). \quad (10)$$

We first reconstruct the part of the Hadasz–Jaskolski proof [1] that has to be modified. Let f be the auxiliary Majorana fermion and ψ the super-Liouville fermion. With their normalization,

$$\chi = f - i\psi, \quad |P, j\rangle = \chi_{-(2j-1)/2} \cdots \chi_{-1/2} |P, 0\rangle \quad (j \in \mathbb{Z}_{>0}). \quad (11)$$

Multiplication by the pole-clearing factor $\Omega(P, j)$ makes every PBW coefficient polynomial in momentum. A degree estimate gives degree at most $j_1^2 + j_2^2 + j_3^2$ (one less for the odd three-form).

The two commuting screening charges are

$$Q_b = \oint dz \psi(z) E^b(z), \quad Q_{1/b} = \oint dz \psi(z) E^{1/b}(z). \quad (12)$$

Reflection of any of the three Fock representations gives eight free-field representatives of the same polynomial. In the unreflected NS vacuum,

$$\langle \chi(z) \chi(w) \rangle = 0, \quad \langle \chi(z) \psi(w) \rangle = -\frac{i}{z - w}. \quad (13)$$

If there are fewer screenings than sea fermions, the numerator therefore vanishes. The eight representatives generate four complete families of linear zeros; the degree bound shows that no momentum-dependent factor is left over.

To determine the remaining constant, choose $N = j_1 + j_2 + j_3$ b -screenings. Every sea fermion must end on a screening, and the fermionic numerator is uniquely

$$(-i)^N \prod_{a=1}^N t_a^{-j_1} (1 - t_a)^{-j_2} \prod_{a < c} (t_a - t_c). \quad (14)$$

The numerator is now an ordinary Selberg integral. The denominator is the Pfaffian Selberg integral evaluated in Appendix B of the paper. Their ratio fixes

$$C_{j_2 j_1}^{j_3}(b) = (-1)^{(j_1 + j_2 - j_3)/2} 2^{(j_1 + j_2 + j_3)/2}. \quad (15)$$

Translating to the human-note momentum and normalization convention gives, for positive n_i ,

$$\begin{aligned} B_a^{\text{NS}}(P_i; n_i) &= \frac{(-1)^{2n_3} (-i)^a}{\prod_{i=1}^3 \sqrt{D_{n_i}(P_i)}} \ell\left(\frac{Q}{2} + P_1 + P_2 + P_3, 2(n_1 + n_2 + n_3)\right) \\ &\quad \times \ell\left(\frac{Q}{2} - P_1 + P_2 + P_3, 2(-n_1 + n_2 + n_3)\right) \\ &\quad \times \ell\left(\frac{Q}{2} + P_1 + P_2 - P_3, 2(n_1 + n_2 - n_3)\right) \\ &\quad \times \ell\left(\frac{Q}{2} - P_1 + P_2 - P_3, 2(-n_1 + n_2 - n_3)\right). \end{aligned} \quad (16)$$

Negative labels are defined by $v_n(P) = v_{-n}(-P)$. This derivation, rather than merely the final product, is what we now generalize.

3 Ramond branch vectors in the free field

The R-sector decomposition proved by Bershtein–Shchepochkin [2] supplies the missing highest vectors. For $n > 0$ put

$$M(n) = 2n - \frac{1}{2} \in \mathbb{Z}_{\geq 0}. \quad (17)$$

The following literal recurrence fixes the two copies and all zero-mode phases. It is the convention used by the code. Start with the tensor ground $|0, 0\rangle$ and the ordered list

$$\mathbf{o}_n = (0, -1, \dots, -M). \quad (18)$$

If $\epsilon \neq M + 1 \pmod{2}$, append a tagged zero 0^* to this list. Read the resulting list from right to left. At each entry r , replace every current tensor state by the two terms

$$|A\rangle_F \otimes |X\rangle_R \mapsto f_r |A\rangle_F \otimes |X\rangle_R - i(-1)^{|A|} |A\rangle_F \otimes \psi_r |X\rangle_R. \quad (19)$$

For the tagged entry only, use $\psi_{0^*} = -\psi_0$ in the second term. The Clifford action is $f_0|a\rangle = 2^{-1/2}|1-a\rangle$ and $\psi_0|g\rangle = 2^{-1/2}|1-g\rangle$, with the usual signs from moving a zero mode through existing creators. The final finite sum is, by definition, $W_n^{(\epsilon)}$. For example,

$$\begin{aligned} W_{1/4}^{(0)} &= |0, 0\rangle + i|1, 1\rangle, \\ W_{1/4}^{(1)} &= 2^{-1/2}(|1, 0\rangle - i|0, 1\rangle). \end{aligned} \quad (20)$$

Thus the one-zero-mode string is the $\epsilon = 1$ copy at $n = 1/4$; the $\epsilon = 0$ copy contains the tagged opposite zero mode. This corrects the potentially ambiguous shorthand “one copy plus the other zero mode.” The states produced by (19) are simultaneous highest vectors of the two Virasoro algebras.

There is an exact finite definition of their norms that does not rely on an all-level product conjecture. Expand the finite string in an auxiliary fermion basis and a Ramond PBW basis,

$$W_n^{(\epsilon)} = \sum_{(A,B,\mu) \in \mathcal{I}_n^{(\epsilon)}} C_{A,B,\mu}^{(\epsilon)}(P, Q) |A\rangle_F \otimes |B, \mu\rangle_R. \quad (21)$$

The coefficients are fixed uniquely by the triangular free-field formulas for L_m, G_r and the chosen G_0 action (26). If G^F and G^R are the finite auxiliary and Ramond Gram matrices at the required level, put

$$\mathcal{N}_R^{(\epsilon)}(n, P) = \sum C_{A,B,\mu}^{(\epsilon)} C_{A',B',\nu}^{(\epsilon)} G_{AA'}^F G_{B\mu,B'\nu}^R. \quad (22)$$

This is a finite rational formula for every fixed n .

The factorized all-level norm suggested by the reflection zeros and the Ramond Whittaker coefficient is

$$\begin{aligned} \|W_n^{(0)}\|^2 &= c_R^{(0)}(n) \frac{\ell(2P, 4n)}{\ell(Q + 2P, 4n)}, & c_R^{(0)}(n) &= 2^{2\lfloor M/2 \rfloor + 1}, \\ \|W_n^{(1)}\|^2 &= c_R^{(1)}(n) \frac{\ell(2P, 4n)}{\ell(Q + 2P, 4n)}, & c_R^{(1)}(n) &= -2^{2\lfloor (M+1)/2 \rfloor}. \end{aligned} \quad (23)$$

The same oscillator argument that proves the NS pole-clearing statement gives the common leg denominator $\delta_n(P)$ in (10). Equation (23) is verified directly for the states needed below, $n = 1/4, 3/4, 5/4$. Its continuation is the natural all-level product conjecture; it is closely related to the

still-conjectural Ramond Whittaker coefficient in Ref. [2]. The exact result below instead uses (22); the compact ℓ -only corollary uses (23).

For the bosonization paragraph only, use canonically normalized complex fermions $\hat{\chi}^\pm = (f \pm i\psi)/\sqrt{2}$. Then

$$\hat{\chi}^\pm = \kappa_\pm : e^{\pm iH} :, \quad \psi = \frac{\hat{\chi}^+ - \hat{\chi}^-}{i\sqrt{2}} \quad (24)$$

which makes the new feature manifest. A screening current is the sum of two H -charges. In the NS calculation (13) forced a unique colour. Between two Ramond spin fields, charge may also run through the Ramond ground doublet. The screened numerator is consequently a sum over colours. This is the precise sense in which the R problem is matrix-valued.

4 The finite Ramond screening kernel

We give a closed definition that involves only finite sums, Pfaffians and gamma functions. It is also an algorithm for arbitrary fixed branch labels.

4.1 Free-fermion part

Move the two Ramond punctures to 0 and ∞ . In either Majorana factor the even Ramond propagator is

$$K_R(z, w) = \frac{z + w}{2\sqrt{zw}(z - w)}. \quad (25)$$

An even matrix element is the Pfaffian of $K_R(z_a, z_b)$. An odd matrix element is the Pfaffian of the augmented antisymmetric matrix whose extra row is $2^{-1/2}z_a^{-1/2}$. The signs of that row are fixed by

$$G_0 w^+ = i\beta e^{-i\pi/4} w^-, \quad G_0 w^- = i\beta e^{i\pi/4} w^+. \quad (26)$$

Here is the mode-extraction convention, which removes another possible ambiguity. Put the punctures at $(\infty, 1, 0)$ and choose local coordinates

$$z_1(x) = x^{-1}, \quad z_2(x) = 1 + x, \quad z_3(x) = x. \quad (27)$$

For a weight-1/2 field X , let $X^{[i]}(x) = (dz_i/dx)^{1/2} X(z_i(x))$, with square-root lifts $(\varsigma_1, \varsigma_2, \varsigma_3) = (-1, i, 1)$. A creator X_{-m} on leg i means the literal coefficient extraction

$$[X_{-m}]_i \Phi = \text{Res}_{x=0} x^{-m-1/2} \Phi(\dots, X^{[i]}(x), \dots). \quad (28)$$

At infinity the string is BPZ reversed and $X_r^\dagger = -X_{-r}$. The auxiliary and physical ground matrices are, respectively,

$$\Gamma_F^0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \Gamma_F^1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \Gamma_\psi^{\eta,0} = \begin{pmatrix} 1 & 0 \\ 0 & \eta \end{pmatrix}, \quad \Gamma_\psi^{\eta,1} = \begin{pmatrix} 0 & 1 \\ i\eta & 0 \end{pmatrix}. \quad (29)$$

For any ordered list of Majorana fields, write $\mathcal{W}_\Gamma$ for Wick contraction using K_R , the augmented row above when the list is odd, and the relevant entry of Γ . Thus no three-point constant is left inside $\mathcal{W}_\Gamma$. For the master component $(\epsilon_3, f) = (0, 0)$ use $\Gamma_F^{a_F}$ with $a_F = 2n_1 + \epsilon_2 \pmod{2}$ and $\Gamma_\psi^{\eta,0}$.

Expand every ordered string (11), (19) by choosing, for each $\chi = f - i\psi$, its auxiliary or physical colour. Let a_i denote such a finite binary path on leg i , $C_i(a_i)$ its sign and power of i , and

$p_F(a_i), p_\psi(a_i)$ its two parities. If $\mathbf{f}(a)$ and $\psi(a)$ are the auxiliary and physical external mode lists, including their ground indices, define the Laurent polynomial by

$$\mathbf{P}_{a_1 a_2 a_3}^{(\eta)}(\mathbf{t}) = \text{ME}_{a_1 a_2 a_3} \left[\mathcal{W}_{\Gamma_F}(\mathbf{f}(a)) \mathcal{W}_{\Gamma_\psi^\eta}(\psi(a), \psi(t_1), \dots, \psi(t_N)) \right], \quad (30)$$

where ME is the ordered product of residues (28). The full fermionic numerator is

$$\mathbf{H}_{\mathbf{n}, \epsilon}^{(\eta)}(\mathbf{t}) = \sum_{a_1, a_2, a_3} (-1)^{p_\psi(a_1)[p_F(a_2)+p_F(a_3)]+p_\psi(a_2)p_F(a_3)} \prod_{i=1}^3 C_i(a_i) \mathbf{P}_{a_1 a_2 a_3}^{(\eta)}(\mathbf{t}). \quad (31)$$

There are no fitted coefficients in (31); its number of summands is finite, at most $2^{j_1+M(n_2)+M(n_3)+2}$ before zero terms are removed. The displayed exponent is increased by one for each tagged opposite zero mode. The η projection is exactly (3).

After pulling out the endpoint powers required by the OPE, the only coincident-point singularity is the simple free-fermion pole. Hence antisymmetry in the screening coordinates implies

$$\mathbf{H}^{(\eta)}(\mathbf{t}) = \prod_a t_a^{-u} (1-t_a)^{-v} \Delta(\mathbf{t})^{-1} F_{\mathbf{n}, \epsilon}^{(\eta)}(\mathbf{t}), \quad \Delta(\mathbf{t}) = \prod_{a < c} (t_a - t_c), \quad (32)$$

where $F^{(\eta)}$ is a symmetric polynomial. In the one-colour NS numerator (14), $F = \Delta^2$, so the effective Vandermonde exponent is shifted upward by two. Direct contractions between the two Ramond strings supply the other terms of the Ramond F .

There is no implicit choice in the endpoint exponents in (32). Given the explicitly extracted Laurent polynomial $\mathbf{H}$, define u to be the negative of its least power in any t_a . After removing $\prod_a t_a^{-u}$ and replacing $t_a = 1 - s_a$, define v in the same way at $s_a = 0$. Fermionic antisymmetry and the simple-pole bound then make $F = \Delta \prod_a t_a^u (1-t_a)^v \mathbf{H}$ a polynomial. This finite valuation prescription can therefore be implemented without making a factorization ansatz for F .

4.2 Evaluation by a finite Jack–Selberg sum

For a symmetric polynomial F define

$$\mathfrak{S}_N[F; A, B, g] = \int_{[0,1]^N} \prod_{a=1}^N dt_a t_a^A (1-t_a)^B |\Delta(\mathbf{t})|^{2g} F(\mathbf{t}). \quad (33)$$

Expand, finitely,

$$F = \sum_{\lambda} f_{\lambda} P_{\lambda}^{(1/g)}, \quad [x]_{\lambda}^{(g)} = \prod_{r=1}^N (x + (1-r)g)_{\lambda_r}. \quad (34)$$

Kadell's Selberg average evaluates (33) as

$$\mathfrak{S}_N[F; A, B, g] = S_N(A, B; g) \sum_{\lambda} f_{\lambda} P_{\lambda}^{(1/g)}(1^N) \frac{[A+1+(N-1)g]_{\lambda}^{(g)}}{[A+B+2+2(N-1)g]_{\lambda}^{(g)}}, \quad (35)$$

where

$$S_N(A, B; g) = \prod_{r=0}^{N-1} \frac{\Gamma(A+1+rg)\Gamma(B+1+rg)\Gamma(1+(r+1)g)}{\Gamma(A+B+2+(N+r-1)g)\Gamma(1+g)}. \quad (36)$$

Thus (35) is a finite sum of explicit gamma and Pochhammer products. It is not an unevaluated contour integral. As in the original Coulomb-gas calculation, the integral is first defined in its convergence chamber and the displayed gamma-function expression gives its meromorphic continuation to the physical value of g .

At a pure b -screening neutrality point

$$\alpha_3 = \alpha_1 + \alpha_2 + Nb, \quad \alpha_i = \frac{Q}{2} + P_i, \quad (37)$$

the bosonic contractions give $A = -b\alpha_1 - u$ and $B = -b\alpha_2 - v$. A second, dual evaluation uses $\alpha_3 = \alpha_1 + \alpha_2 + N/b$ and $b \mapsto b^{-1}$. For the first chamber, the common Pfaffian-Selberg exponent is

$$g = -\frac{bQ}{2} = -\frac{b^2 + 1}{2}. \quad (38)$$

For the one-colour insertion $F = \Delta^2$, this becomes the ordinary Selberg exponent $g + 1 = (1 - b^2)/2$, reproducing the numerator integral used by Hadasz–Jaskolski. The numerator of the normalized screened matrix element is therefore (35) with the polynomial obtained from (31). Its denominator is the same expression with all three branch strings removed but with the same R ground component. More explicitly, let $\mathbf{g}_{\epsilon_2\epsilon_3;f}^{(\eta)}$ be the raw charge-basis ground amplitude for the selected component. If $F_0^{(\eta,f)}$ and (u_0, v_0) denote the corresponding ground polynomial and its endpoint valuations, then at the neutrality point (37) we define

$$\mathcal{R}_{\epsilon_2\epsilon_3;f}^{(\eta)}|_N = \mathbf{g}_{\epsilon_2\epsilon_3;f}^{(\eta)} \delta_{n_1}(P_1) \delta_{n_2}(P_2) \delta_{n_3}(P_3) \frac{\mathfrak{S}_N[F_{\mathbf{n},\epsilon}^{(\eta)}; -b\alpha_1 - u, -b\alpha_2 - v, g]}{\mathfrak{S}_N[F_0^{(\eta,f)}; -b\alpha_1 - u_0, -b\alpha_2 - v_0, g]}. \quad (39)$$

The ordered-contour factorial is common to numerator and denominator and cancels in this ratio. The prefactor restores the ground normalization removed by forming the ratio; in particular $\mathbf{g}_{00;0}^{(\eta)} = 1 + i\eta$ and $\mathbf{g}_{10;0}^{(\eta)} = -e^{-i\eta\pi/4}$ in the convention (26). We use the shorter notation

$$\mathcal{R}_{\epsilon_2\epsilon_3;f}^{(\eta)}(P_1, P_2, P_3; n_1, n_2, n_3). \quad (40)$$

Equations (31)–(39) give a closed finite evaluation of the native ground-resolved Coulomb functional. It is a fixed SCA chiral form only when the screening holonomy selects that form. Applying a constant ground-space rotation to the other excited component is not justified; the reflected vertex must be evaluated independently.

For completeness, a conditional screening reconstruction away from a neutrality hyperplane follows if the pole-cleared matrix element is a polynomial with the following sharp degree. For the even component put

$$d = j_1^2 + j_2^2 + j_3^2 - \frac{1}{2}, \quad (41)$$

and the odd component has degree at most $d - 1$. Define N_0 to be the smallest element of $\{0, 1\}$ for which the ground denominator in (39) is nonzero for the selected component. Choose $d + 1$ neutrality points of that parity, $N = N_0, N_0 + 2, \dots, N_0 + 2d$. If their third momenta are $P_3^{(a)}$, then

$$\mathcal{R}(P_3) = \sum_{a=0}^d \mathcal{R}(P_3^{(a)}) \prod_{\substack{c=0 \\ c \neq a}}^d \frac{P_3 - P_3^{(c)}}{P_3^{(a)} - P_3^{(c)}}. \quad (42)$$

If every node in (42) is independently evaluated as the same analytic SCA form, this would define it at generic (P_1, P_2, P_3) . That premise is currently missing for the complementary Ramond charge chart: the reflected fermion contains bosonic-current modes. The degree bound is supported by clearing the reflection denominators in the finite PBW oracle—leg i contributes j_i^2 , while the two Ramond zero-mode seas remove one half in total—but a degree bound does not supply the absent screening nodes.

The logical status should be stated precisely. Polynomiality and this degree argument are the all-level step needed to promote the screened values (39) to the generic value (42). They are proved in the NS case and verified for the R states used in Section 6; a general R proof is not supplied by the cited literature. Hence the screening-only interpolation is a concrete all-level conjecture, not by itself the unconditional definition below.

4.3 Generic-momentum PBW/Ward oracle

For comparison we now give a finite evaluator with no screening number or interpolation nodes. This is the object against which the contour calculation is tested; it is not used to derive a screened value. At each physical level let $T_\sigma(P, Q)$ be the square matrix whose columns are SCA PBW words expanded in the free Fock basis using, for negative modes,

$$\begin{aligned} L_k^{(\sigma)} &= \frac{1}{2} \sum_{m \neq 0, k} : c_{k-m} c_m : + \frac{1}{2} \sum_r r : \psi_{k-r} \psi_r : + \frac{i}{2} (kQ + 2\sigma P) c_k, \\ G_r^{(\sigma)} &= \sum_{m \neq 0} c_m \psi_{r-m} + i(rQ + \sigma P) \psi_r. \end{aligned} \quad (43)$$

For positive branch labels $\sigma = -1$; reflection gives the other sheet. Expand the NS sea (11) and the R recurrence (19), and multiply every physical Fock endpoint by T_-^{-1} . Denote the resulting finite component lists by $\mathcal{C}_i$. An element $x_i \in \mathcal{C}_i$ consists of an auxiliary Fock state A_i , a physical PBW state V_i , and its rational coefficient $c_i(P_i, Q)$.

Let $\rho_F^{a_F}$ be the auxiliary Majorana three-form fixed by (25), (28), and $\Gamma_F^{a_F}$. Let $\rho_{\text{SCA}}^{\eta, 0}$ be the unique NS–R–R SCA three-form obtained from the SCA commutators and the ground matrix $\Gamma_\psi^{\eta, 0}$. Put

$$\begin{aligned} a_F &= 2n_1 + \epsilon_2 \pmod{2}, & \eta_{\text{eff}} &= (-1)^{2n_1} \eta, \\ \mathcal{A}_{\epsilon_2}^{(\eta)}(b, \mathbf{P}; \mathbf{n}) &= \sum_{x_i \in \mathcal{C}_i} (-1)^{\kappa(x_1, x_2, x_3)} c_1 c_2 c_3 \rho_F^{a_F}(A_1, A_2, A_3) \rho_{\text{SCA}}^{\eta_{\text{eff}}, 0}(V_1, V_2, V_3), \\ \kappa &= p_\psi(x_1)[p_F(x_2) + p_F(x_3)] + p_\psi(x_2)p_F(x_3) \pmod{2}. \end{aligned} \quad (44)$$

All three sums in (44) are finite, so this is an unambiguous PBW/Ward evaluator for every fixed set of labels; it is implemented in `Code/screening_kernel.py`. It is nevertheless tautological from the viewpoint of the desired free-field calculation, because ρ_{SCA} is left to the Ward recursion. The pole-cleared notation used above is only the abbreviation

$$\mathcal{R}_{\epsilon_2 0; 0}^{(\eta)} = \delta_{n_1}(P_1) \delta_{n_2}(P_2) \delta_{n_3}(P_3) \mathcal{A}_{\epsilon_2}^{(\eta)}. \quad (45)$$

Equation (39) is the proposed screening-charge evaluation of this already-defined function on a neutrality hyperplane.

4.4 Normalized branching coefficient

It is convenient to present the master component with $\epsilon_3 = f = 0$ first. Let

$$c_{\text{NS}}(n) = (-1)^{2n} 2^{2n}, \quad \mathcal{N}_{\text{NS}}(n, P) = c_{\text{NS}}(n) \frac{\ell(2P, 4n)}{\ell(Q + 2P, 4n)}, \quad (46)$$

and use the exact finite R norm (22). In the raw charge-basis normalization of (11) and (19), the PBW/Ward oracle for the matrix element is

$$\hat{\rho}_0^{(\eta)} = \mathcal{A}_{\epsilon_2}^{(\eta)} = \frac{\mathcal{R}_{\epsilon_2 0; 0}^{(\eta)}}{\delta_{n_1}(P_1) \delta_{n_2}(P_2) \delta_{n_3}(P_3)}. \quad (47)$$

Combining (47) with the three raw norms gives the exact normalization identity

$$\left(\mathbb{B}_f^{(\eta)}[\mathbf{n}; \epsilon_2, \epsilon_3] \right)^2 = (-1)^{\epsilon_3} (-i)^f \frac{\left(\mathcal{A}_{\epsilon_2}^{(\eta)} \right)^2}{\mathcal{N}_{\text{NS}}(n_1, P_1) \mathcal{N}_R^{(\epsilon_2)}(n_2, P_2) \mathcal{N}_R^{(0)}(n_3, P_3)}. \quad (48)$$

Here $\mathcal{A}$ is the PBW/Ward sum (44). Equation (48) becomes a free-field result only after the same raw matrix element has been obtained from the screening contour.

If the factorized norm conjecture (23) is used, (48) reduces to the compact ℓ -only expression

$$\left(\mathbb{B}_f^{(\eta)}[\mathbf{n}; \epsilon_2, \epsilon_3] \right)^2 = (-1)^{\epsilon_3} (-i)^f \frac{2^{1/2} \left(\mathcal{R}_{\epsilon_2 0; 0}^{(\eta)} \right)^2}{c_{\text{NS}}(n_1) c_R^{(\epsilon_2)}(n_2) c_R^{(0)}(n_3) \prod_{i=1}^3 D_{n_i}(P_i)}. \quad (49)$$

Under both conjectural replacements just stated—the factorized R norm and the generic-momentum screening interpolation—all the continuous dependence on b, P_i, n_i in (49) lies in the finite polynomial $\mathcal{R}$ defined by (31)–(42). The ϵ_3 and f dependence in (48) is universal. The ϵ_2 dependence is retained inside the screening polynomial; it is not legitimate to discard it before doing the zero-mode projection. Negative branch labels require no new formula. Reflect one leg at a time,

$$(n_i, P_i) \mapsto (-n_i, -P_i), \quad (50)$$

use (9), and evaluate the positive representative. The possible sign of the reflected highest vector disappears from the squared coefficient (48).

5 Relation to the four- ℓ conjecture

Define the scalar Hadasz–Jaskolski continuation

$$\begin{aligned} \mathcal{P}(\mathbf{n}; \mathbf{P}) &= \ell\left(\frac{Q}{2} + P_1 + P_2 + P_3, 2(n_1 + n_2 + n_3)\right) \\ &\quad \times \ell\left(\frac{Q}{2} - P_1 + P_2 + P_3, 2(-n_1 + n_2 + n_3)\right) \\ &\quad \times \ell\left(\frac{Q}{2} + P_1 + P_2 - P_3, 2(n_1 + n_2 - n_3)\right) \\ &\quad \times \ell\left(\frac{Q}{2} - P_1 + P_2 - P_3, 2(-n_1 + n_2 - n_3)\right). \end{aligned} \quad (51)$$

The quantity that should be divided out, rather than subtracted, is

$$U_{\epsilon_2}^{(\eta)} = \frac{\mathcal{R}_{\epsilon_2 0; 0}^{(\eta)}}{\mathcal{R}_{\epsilon_2 0; 0}^{(\eta)}|_{F=\Delta^2}} = \frac{\mathcal{R}_{\epsilon_2 0; 0}^{(\eta)}}{\text{the appropriate reflected-sheet version of } \mathcal{P}}. \quad (52)$$

Here $F = \Delta^2$ means retaining only the path in which every external sea fermion ends on a screening, as in (14). In the NS sector $U = 1$ identically. In the R sector it remains one whenever the charge projection forces a single screening colour. In general U is a momentum-dependent finite Jack sum. It is not the field normalization $\Omega_k(\alpha)$ of the double-Liouville dictionary [3], and it is not a momentum-independent normalization constant.

6 Low-level channels and direct screening restrictions

This section is placed after the derivation deliberately.

6.1 Two Ramond ground branches

For $(n_1, n_2, n_3) = (0, 1/4, 1/4)$ the symmetric screening polynomial is constant. In the raw convention,

$$\mathcal{R}_{00;0}^{(\eta)} = 1 + i\eta, \quad \mathcal{R}_{10;0}^{(\eta)} = -e^{-i\eta\pi/4}. \quad (53)$$

Using $c_R^{(0)}(1/4) = 2$ and $c_R^{(1)}(1/4) = -1$ in (49) gives, for both ϵ_2 ,

$$\left(\mathbb{B}_f^{(\eta)}\left[0, \frac{1}{4}, \frac{1}{4}; \epsilon_2, \epsilon_3\right]\right)^2 = (-1)^{\epsilon_3}(-i)^f \frac{i\eta}{2}. \quad (54)$$

6.2 First crossed kernel

The first nonconstant F occurs when both Ramond strings contain a nonzero mode, at $(0, 3/4, 3/4)$. Put

$$x_{++} = \frac{Q}{2} + P_1 + P_2 + P_3, \quad x_{--} = \frac{Q}{2} + P_1 - P_2 - P_3, \quad (55)$$

$$x_2 = \frac{Q}{2} - P_1 + P_2 + P_3, \quad E_j = Q + 2P_j, \quad (56)$$

$$a_j = (2P_j)^2 + Q(2P_j) + 1, \quad d_j = E_j^2 + QE_j + 1, \quad (57)$$

$$K = (x_{++}^2 + Qx_{++} + 1)(x_2^2 + Qx_2 + 1), \quad L = x_{++}(x_{--} - Q), \quad (58)$$

$$H = L^2 + 2L(E_2E_3 + 1) + d_2d_3. \quad (59)$$

The one-colour component gives K . The remaining contraction paths in $F^{(-)}$ give the crossed result H , equivalently

$$H = (1, L) \begin{pmatrix} d_2d_3 & 1 + E_2E_3 \\ 1 + E_2E_3 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ L \end{pmatrix}. \quad (60)$$

For the $\epsilon_2 = 0$ master the normalized answer is

$$\left(\mathbb{B}_f^{(\eta)}\left[0, \frac{3}{4}, \frac{3}{4}; 0, \epsilon_3\right]\right)^2 = (-1)^{\epsilon_3}(-i)^f \frac{i\eta}{2} \frac{F_\eta^2}{a_2a_3d_2d_3}, \quad F_+ = K, \quad F_- = H. \quad (61)$$

The $\epsilon_2 = 1$ component is obtained from the same literal sum (44); it is not implicitly identified with (61). Since H is irreducible over the rational function field in (Q, P_1, P_2, P_3) , it cannot be a four- ℓ product on any reflected momentum sheet. This is not an obstruction to free fields; it is the first place where the two-colour screening polynomial is nonconstant. Relative to the literal four- ℓ factor, $U_+ = 1$ and $U_- = H/K$.

6.3 Direct screening matrix element on two hyperplanes

We now compute rather than reconstruct the first crossed three-form. Keep the puncture order $(\infty, 1, 0) = (\text{NS}_1, R_2, R_3)$ and define

$$\mathfrak{M}_f^{(\eta)} = \frac{\langle v_{1,0}(P_1) | V[v_{2,3/4}^0(P_2)](1) | v_{3,3/4}^0(P_3) \rangle_f^{(\eta)}}{\langle v_{1,0}(P_1) | V[w_2](1) | w_3 \rangle_f^{(\eta)}}. \quad (62)$$

The numerator uses the literal consecutive $\chi = f - i\psi$ strings on the two Ramond fields and N insertions of $\psi(t)E^b(t)$. The denominator is the primary two-spin BFL integral in the same ground channel.

At $N = 2$, charge neutrality is

$$P_1 = -\frac{Q}{2} - P_2 - P_3 - 2b. \quad (63)$$

Writing $e_1 = t_1 + t_2$ and $e_2 = t_1 t_2$, direct mode extraction from the two χ strings and the two-spin Pfaffian gives, for $f = 0$ and $\eta = -1$,

$$F_2(t_1, t_2) = \frac{1-i}{2} (2e_2 - e_1) (e_2^2 - e_1 e_2 + e_1^2 - 3e_2). \quad (64)$$

Expanding only in products of elementary symmetric functions and applying their exact Selberg moments yields

$$\begin{aligned} \mathfrak{M}_0^{(-)} &= (-1+i)R_2, & \mathfrak{M}_1^{(-)} &= -\sqrt{2}iR_2, \\ R_2 &= \frac{13b^4 + 10b^3(P_2 + P_3) + 4b^2P_2P_3 + 8b^2 + 4b(P_2 + P_3) + 4}{(b^2 + 2bP_2 + 2)(b^2 + 2bP_3 + 2)}. \end{aligned} \quad (65)$$

This computation imports no SCA three-form. Symbolic reduction gives the nontrivial identity

$$R_2 = \left. \frac{H}{d_2 d_3} \right|_{(63)}. \quad (66)$$

Thus the irreducible H is directly visible in the screening integral, not only in the PBW calculation.

At the natural $N = 3$ node, $P_1 = -Q/2 - P_2 - P_3 - 3b$, the physical form is $\eta = +1$. The external-screen block reduces the Pfaffian to one determinant, followed by an ordinary Selberg product. The answer is

$$\begin{aligned} \mathfrak{M}_0^{(+)} &= -4(1+i)R_3, & \mathfrak{M}_1^{(+)} &= -4\sqrt{2}R_3, \\ R_3 &= \frac{b^2(3b^2 - 1)(2b^2 + bP_2 + bP_3 + 1)(5b^2 + 2bP_2 + 2bP_3 + 1)}{(b^2 + 2bP_2 + 2)(b^2 + 2bP_3 + 2)(2b^2 + 2bP_2 + 1)(2b^2 + 2bP_3 + 1)}. \end{aligned} \quad (67)$$

Direct simplification gives

$$4R_3 = \left. \frac{K}{d_2 d_3} \right|_{P_1 = -Q/2 - P_2 - P_3 - 3b}. \quad (68)$$

Equations (66) and (68) are genuine contour checks of the two channels. They do not yet determine either generic degree-four polynomial from screening data alone. The executable derivation is `Code/direct_matrix_element_trial.py`.

7 Independent PBW/Ward comparison through onset level 7/2

The comparison uses explicit simultaneous $\text{Vir} \oplus \text{Vir}$ primaries and the generalized (NS, R, R) Ward identities, not the screening contours. The intrinsic NS-primary parity is mandatory: the Ward sign uses

$$\eta_{\text{eff}} = (-1)^{p_\phi} \eta, \quad p_\phi = 2n_1 \pmod{2}. \quad (69)$$

The labels whose individual branch onset does not exceed 7/2 are

$$n_1 \in \{0, \frac{1}{2}, 1\}, \quad n_2, n_3 \in \{\frac{1}{4}, \frac{3}{4}, \frac{5}{4}\}. \quad (70)$$

There are 27 level triples, four independent continuous masters (ϵ_2, η) per triple, and sixteen discrete restrictions after (ϵ_3, f) are restored. The audit checks the following separately:

1. the raw R norms at $n = 1/4, 3/4, 5/4$;
2. all four PBW/path master amplitudes at each of the 27 triples;
3. the universal reduction $(\mathbb{B}_f^{(\eta)})^2 = (-1)^{\epsilon_3} (-i)^f (\mathbb{B}_0^{(\eta)}[\epsilon_2, 0])^2$;
4. the scalar-product specialization and its first failure (61).

The exact arithmetic report and the commands used to regenerate it are in `Code/verification_report.md`. The continuous grid audit constructs the branch strings in two different ways (finite colour paths versus direct simultaneous-primary assembly); both use the same independently audited Ward functional. It is therefore a check of the literal finite function, normalization and phase restoration, not a proof of the generic-momentum screening interpolation or of the Jack–Selberg identity. The comparison confirms (44) throughout (70). The scalar four- ℓ specialization matches only the one-colour components; the crossed components are required already at $(0, 3/4, 3/4)$.

8 Conclusion

The correct primary object is the free-field matrix element (62). For the first crossed branch its two-screening evaluation gives the irreducible H restriction (66), and its natural three-screening evaluation gives the factorized K restriction (68). These are genuine Pfaffian–Selberg calculations and show directly that the factor left after the four- ℓ term is removed can be momentum dependent.

The finite function (44) remains useful as a PBW/Ward oracle and has passed the stated low-level comparisons, but it is not a derivation of the contour integral.

The status of the residue method is more precise. In an ordinary scalar Selberg family, endpoint support fixes the Gamma divisor and a collision residue fixes the remaining normalization from the $N = 0, 1$ base cases. The Ramond descendant integral is vector-valued, and the two endpoint residues proved here contain complementary rank-one projectors. Neither relation alone determines the full function. A generic all-level answer requires a proof of finite contour-orbit closure, degree and pole bounds on that space, a full-rank collection of endpoint/cluster maps, and the inversion/crossing connection matrix. These data must include the current–fermion terms of the reflected Ramond vertex and have no common invisible subspace. Once the vector Selberg integral is fixed from its base vectors, enough same-parity screening nodes can separately reconstruct the pole-cleared generic-momentum function by finite interpolation.

Consequently this note proposes the multi-channel screening organization and proves its first nontrivial restrictions; it does not claim a completed generic-momentum Ramond branching formula. The projectors are auxiliary data in an evaluation algorithm, not additional physical branching coefficients.

References

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